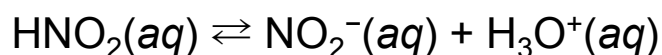


Assume HNO_2 dissociates in water with a K_a of 4×10^{-5} at 298 K.



An aqueous solution containing a mixture of HNO_2 and NO_2^- has a pH of 6.2 at 298 K. Compared to the concentration of NO_2^- , the concentration of HNO_2 must be:

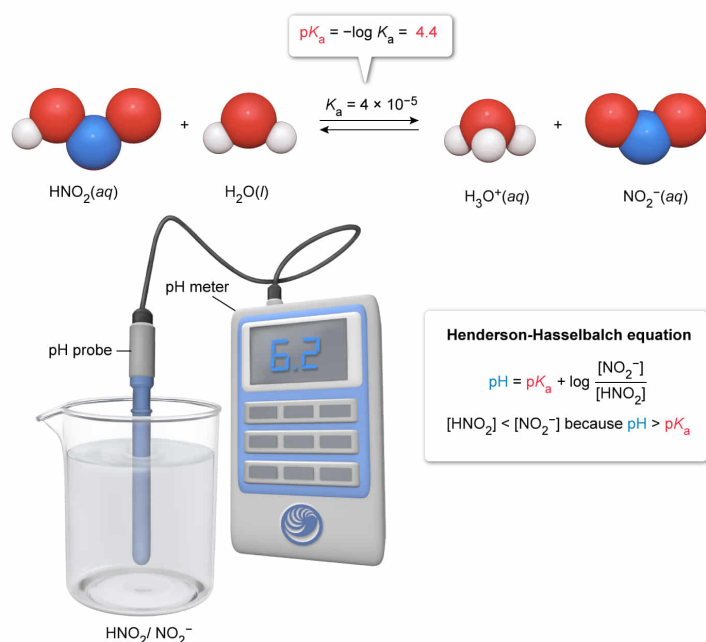
- ☐ A. greater, because the pH of the solution is greater than its $\text{p}K_a$.
[28%]
- ✓ ☒ B. less, because the pH of the solution is greater than its $\text{p}K_a$.
[52%]
- ☐ C. greater, because the pH of the solution is less than its $\text{p}K_a$.
[12%]
- ☐ D. less, because the pH of the solution is less than its $\text{p}K_a$. [5%]

Correct

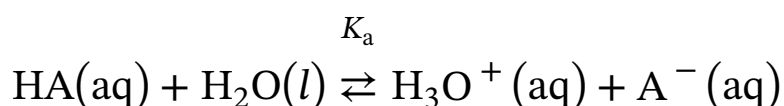
52% answered correctly

Explanation:

The relationship between pH, $\text{p}K_a$, and concentration



The dissociation of a **weak acid** (HA) in water can be represented by the general equilibrium reaction:



The extent to which an **acid** dissociates in water is characterized by the **acid dissociation constant** K_a , which is also expressed as the $\text{p}K_a$:

$$\text{p}K_a = -\log K_a$$

Without a calculator, the negative logarithm can be approximated using the relationship:

$$-\log(m \times 10^{-n}) \approx n - 0.m$$

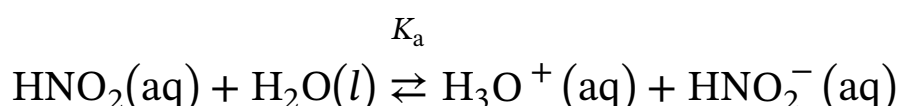
where n is a whole number and m is a number between 1 and 10.

A solution's acidity is measured by its **pH**, which depends on the **molar concentration** of H_3O^+ . The relationship between $\text{p}K_a$ and pH can be quantitatively described by the **Henderson-Hasselbalch equation**:

$$\text{pH} = \text{p}K_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

If the **pH** of a solution is **less than** the **$\text{p}K_a$** (ie, the solution is more acidic), the equilibrium is shifted toward a greater concentration of the **protonated** acid (ie, $[\text{HA}] > [\text{A}^-]$). Conversely, if the **pH** is **greater than** the **$\text{p}K_a$** (ie, the solution is more basic), the equilibrium is shifted toward a greater concentration of the deprotonated **conjugate base** (ie, $[\text{HA}] < [\text{A}^-]$).

The dissociation of HNO_2 in water ($K_a = 4 \times 10^{-5}$) proceeds according to the equilibrium reaction:



As such, the pK_a is:

$$pK_a = -\log (4 \times 10^{-5}) = 4.4$$

Without a calculator, the pK_a of HNO_2 is approximated as:

$$pK_a \approx 5 - 0.4 = 4.6 \approx 4.4$$

The question states that an aqueous solution containing a mixture of HNO_2 and NO_2^- (ie, a **buffer**) has a pH of 6.2. Therefore, the **concentration of HNO_2 must be less than NO_2^-** (ie, $[\text{HNO}_2] < [\text{NO}_2^-]$) because the **pH of the solution is greater** than the **pK_a** (ie, $6.2 > 4.4$).

(Choices A, C, and D) The pH of the solution is *greater than* the pK_a of HNO_2 ; consequently, *less* HNO_2 is present in the solution compared to NO_2^- .

Educational objective:

The pK_a is a measure of the acidity of a particular proton in a molecule.

In a solution containing a weak acid (HA) and its conjugate base (A^-), when the pH is greater than the pK_a , the amount HA is less than the amount of A^- and vice versa.

Time spent:0 sec

Subject:
General Chemistry

QID:404004

System:
Solutions and Electrochemistry